NeuroSTEM

Computing of Neuroscience Data

Companion quiz

Name: _____ Date: _____

Write short answers. Use the workshop atlas to help with question 3.

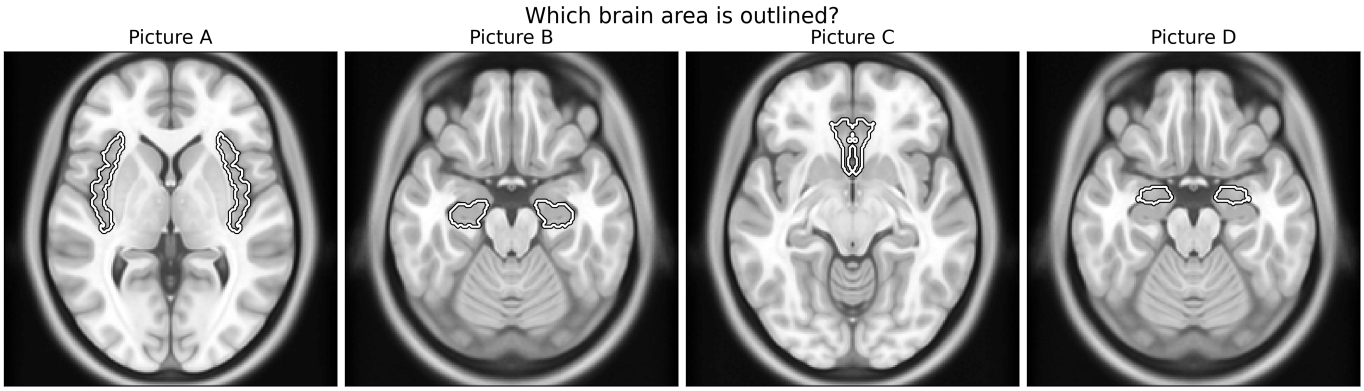
1. What is one example of what MRI is useful for?

2. What is one example of what PET is useful for?

3. Name the brain area outlined in each picture.

Use each name once: Amygdala, Anterior cingulate, Hippocampus, Insula.

The pictures show different slice levels through the brain.



A: _____ **B:** _____

C: _____ **D:** _____

4. Why do we line up (register) the MRI and PET images before measuring PET signal in a brain area?

5. If stress ratings and PET signal are related in a group, does that prove that stress caused the PET change? Why?

Workshop: https://ashgillman.github.io/neurostem-2026/

Teaching examples: workshop PET signals and participant data are simulated.

MRI: MNI152 (Fonov et al., 2011); atlas: CerebrA (Manera et al., 2020), via TemplateFlow.

Worksheet: Ashley Gillman, CC BY 4.0. Image terms and notices: DATA_SOURCES.md / NOTICE.

MRI template: Copyright (C) 1993-2004 Louis Collins, McConnell Brain Imaging Centre, Montreal Neurological Institute, McGill University.